TECHNICAL REPORT

Sixteen Rounds of Adversarial Hardening Against an NVML-Telemetry Training Detector

Round-by-round results from red-teaming a from-scratch reimplementation of Rahman & Tajdari's train-vs-infer classifier on real Azure A100 hardware.

8×A100 Azure ML Compute Instance, GPUs 3–5 Real hardware execution, no simulation
Full data: docs/REDTEAM.md

ABSTRACT

We red-team a RandomForest classifier that infers whether a GPU is training or serving inference from per-GPU power/memory/utilization telemetry alone — no code, weights, or network access. Across 16 rounds of an iterative hardening loop (attempt an evasion, fold the caught example into training, retest a fresh execution), every mechanism tried — communication-shape disguises, memory-footprint reduction, cross-mechanism transfer, and a combined adversary — was ultimately caught, with one axis (memory-footprint reduction) coming within reach of a genuine evasion (P(training)=0.588, round 8) before the classifier adapted. Two rounds (10–11) and one combined attempt (14) produced confounded, low-power out-of-distribution readings that should not be read as evidence of evasion difficulty either way — a methodological caution that, on later audit while preparing this report, we found had been under-applied to one earlier round (2) as well.

1. Method

A RandomForest (400 trees, class-weight balanced — matching Rahman & Tajdari's own model choice) trained on 12 window-level features: structural (mean power, mean utilization), temporal (coefficient of variation and autocorrelation of power/utilization, spectral periodicity), and three physics-based features from the paper's own hardened round: cumulative energy, power–memory correlation, and early memory-allocation timing. Telemetry is sampled at 1 Hz via `pynvml` from a small (~85–100M parameter) transformer running real distributed workloads — data-parallel and tensor-parallel inference, DiLoCo-style low-communication training, and five purpose-built adversarial variants.

Protocol. Each round attempts a new evasion strategy, is scored against the classifier hardened on every *prior* round only, and — regardless of outcome — is then folded into training as a correctly-labeled positive example before the next round runs. This mirrors Rahman & Tajdari's own §4.3 co-evolution methodology. Fresh executions are always used for testing; a round is never evaluated on data just folded into its own training set.

2. Results

2.1 Communication-shape evasion (rounds 0–7)

Reshaping how gradients synchronize — chunked AllReduce disguised as KV-cache traffic, DiLoCo's low-frequency outer loop, and both combined — evaded detection exactly once, on the undefended baseline (round 0). After a single fold-in, every subsequent variant of this idea was caught, with confidence trending upward as the classifier accumulated examples of "unusual communication shape."

RD	MECHANISM	P(TRAIN)	NOTE
0	kv_disguise, chunk=2048	0.10	evaded
1	diloco, inner=25	0.84	1st fold-in of R0
2	kv_disguise, chunk=512	0.80	† see §4 errata
3	kv_disguise, chunk=8192	0.64	—

RD	MECHANISM	P(TRAIN)	NOTE
4	diloco, inner=10	0.85	—
5	diloco, inner=100	0.96	—
6	composite, 1024/20	1.00	2 mechanisms stacked
7	composite, 4096/50	1.00	—

Root cause, checked against the raw data rather than assumed: raw power does not separate the classes (inference reads slightly *higher* mean power than training in this data). The generalizing signal is memory — AdamW's momentum/variance state produces a first-30-second memory ramp (5,480–10,392 MB across training variants) that no communication-shape change touches, versus 1,528–3,366 MB for inference. One example teaches the classifier this once; every later communication-only disguise inherits the same tell for free.

2.2 Memory-footprint evasion (rounds 8–11)

The only axis that ever produced a near-evasion. Replacing AdamW with momentum-free SGD removes persistent optimizer state entirely.

RD	MECHANISM	P(TRAIN)	NOTE
8	SGD, momentum=0, b=4/s=512	**0.59**	near coin-flip closest evasion overall
9	R8 + chunked comms	1.00	combining backfired
10	SGD, b=1/s=128	0.67	⚠ 135.0 W, out-of-band
11	SGD, b=2/s=256	0.95	⚠ 208.1 W, out-of-band

Round 8's memory delta (3,968 MB) moved close to inference's own baseline (3,366 MB), and confidence dropped to 0.59 — real progress on the targeted feature. Layering communication disguise back on top (round 9) re-triggered the already-learned "unusual comm shape" tell and pushed confidence back to 1.00: the classifier's lesson from §2.1 generalized against this axis too. Rounds 10–11 pushed batch/sequence length further to shrink activation memory as well — confidence rose, non-monotonically, rather than falling. Mean power also dropped to 133–208 W, below the 239–351 W band the original benign+adversarial baseline run established. A classifier's prediction on an input this far outside its training distribution is not reliable evidence in either direction; we read rounds 10–11 as inconclusive on the evasion question, not as "harder to evade."

2.3 Cross-mechanism transfer (rounds 12–13)

To test whether "infrequent synchronization" was a genuine feature-level pattern or a DiLoCo-specific fingerprint, gradient accumulation — a structurally unrelated technique with no outer-loop momentum state — was used to produce the same external communication-frequency reduction.

RD	MECHANISM	P(TRAIN)	NOTE
12	grad. accumulation, every 4 steps	1.00	higher confidence than DiLoCo itself
13	grad. accumulation, every 25 steps	1.00	—

Caught at higher confidence than any DiLoCo round — evidence the classifier generalized a mechanism-agnostic pattern rather than memorizing one library's traffic shape.

2.4 Combined adversary — a confounded result (round 14)

Combined every partial win so far — SGD/momentum=0, realistic batch/sequence length, no communication changes — plus a new axis: deliberate idle gaps between steps, targeting temporal/duty-cycle features no prior round had touched.

RD	MECHANISM	P(TRAIN)	NOTE
14	R8 + idle-gap duty cycling	1.00	⚠ 163.6 W, out-of-band

Worse than memory-only round 8, not better. The temporal-shaping idea worked on its own terms (power variance rose as intended), but idle sleeping pulled mean power to 163.6 W — again outside the reference band. We cannot distinguish "temporal shaping doesn't help" from "this input again left the region the classifier was trained on." Treated as inconclusive, not a negative result.

2.5 Memory evasion, retested without the confound (round 15)

Gradient checkpointing (trading compute for activation memory) at realistic parameters, with no communication or temporal changes — designed specifically to avoid round 14's power confound.

RD	MECHANISM	P(TRAIN)	NOTE
15	gradient checkpointing, b=4/s=512	0.78	power in-band; larger memory reduction than R8

Power held at 319 W (in-band) and memory delta fell further than round 8's (2,432 vs 3,968 MB) — the cleanest memory-axis result in the campaign. Confidence was still higher than round 8's, most likely because the classifier had by this point seen four prior memory-focused rounds: novelty of the axis, not magnitude of the change on one feature, appears to be what determines round difficulty.

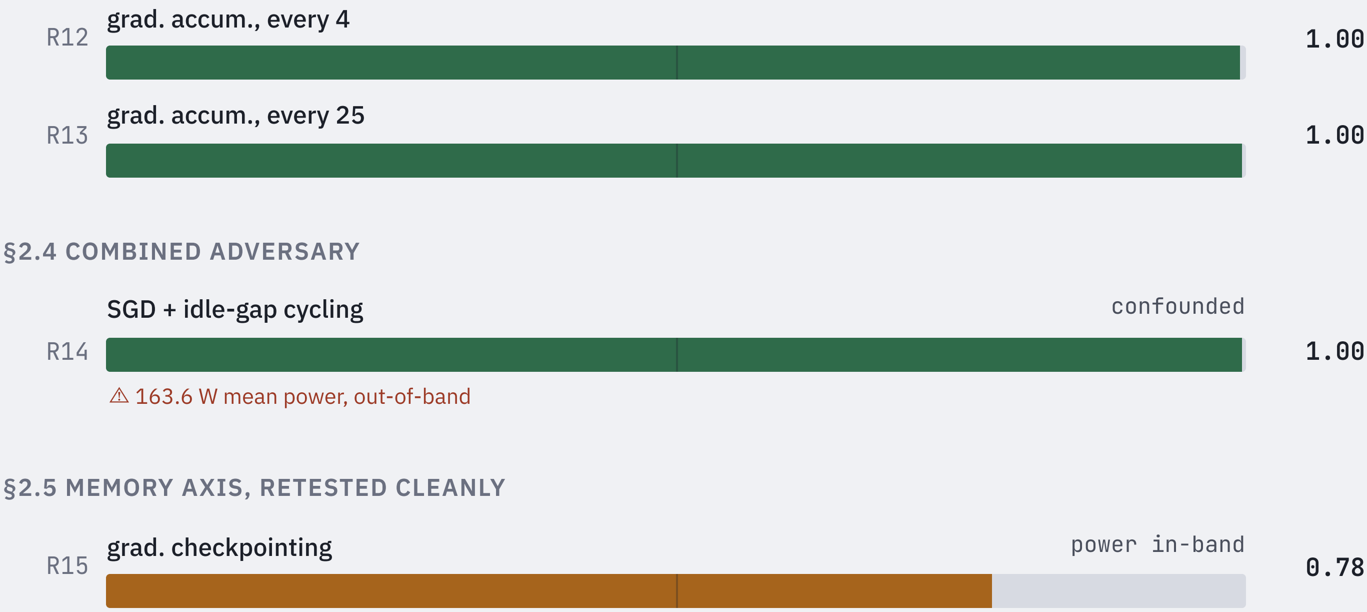
3. Figure 1 — confidence trajectory across all 16 rounds

P(training) per round, grouped by evasion axis. Reference threshold at 0.5.

caught with high confidence (≥0.85) | caught, lower confidence (0.5–0.85) | at or below threshold (<0.5)

⚠ power outside 239–351 W reference band — reading unreliable

§2.3 CROSS-MECHANISM TRANSFER



4. Errata

CORRECTION, ADDED WHILE PREPARING THIS REPORT

The original round-by-round writeup (docs/REDTEAM.md) stated that mean power for rounds 10–11 and 14 fell "outside the 240–351 W range every real category has ever shown," implying that band held for every round in the campaign. Recomputing power for all 16 rounds to build Figure 1 showed this was not quite right: round 2 (kv_disguise, chunk=512) also reads at 170.8 W, outside that band, likely because its unusually fine chunking made the run communication-bound enough to lower average power throughout — and it was never flagged at the time. The 239–351 W band is accurately described as the range the *original benign+adversarial baseline run* established, not a literal floor for every round ever executed. docs/REDTEAM.md has been corrected in three places; this report states the band's provenance precisely from the outset.

5. Table 1 — full round data

RD	MECHANISM	P(TRAIN)	POWER (W)	MEM Δ (MB)
R0	kv_disguise chunk=2048	0.104	278.1	5480
R1	diloco inner=25	0.842	318.0	5924
R2	kv_disguise chunk=512	0.799	170.8 ⚠	5480

RD	MECHANISM	P(TRAIN)	POWER (W)	MEM Δ (MB)
R3	kv_disguise chunk=8192	0.640	309.2	5480
R4	diloco inner=10	0.850	316.9	5924
R5	diloco inner=100	0.959	319.3	5924
R6	composite 1024/20	1.000	302.3	5926
R7	composite 4096/50	0.998	316.3	5924
R8	SGD momentum=0	0.588	315.7	3968
R9	R8 + chunked comms	0.999	278.6	4808
R10	SGD b=1/s=128	0.675	135.0 ⚠	1904
R11	SGD b=2/s=256	0.947	208.1 ⚠	2080
R12	grad. accum. every-4	0.996	323.3	4470
R13	grad. accum. every-25	0.997	326.8	4470
R14	SGD + idle-gap cycling	0.998	163.6 ⚠	3968
R15	gradient checkpointing	0.777	319.3	2432

Reference band for mean power, from the original benign+adversarial baseline: 239–351 W.

6. Discussion

- **Communication-shape disguises are self-defeating once caught once.** The classifier does not learn to recognize a specific trick — it learns that *any* unusual synchronization pattern is itself a positive signal, so further disguising makes detection more confident, not less (§2.1, §2.3).
- **Memory footprint is the one axis with real, if partial, traction.** Removing optimizer state closed most of the gap to inference on the targeted feature and produced the campaign's only near-coin-flip confidence score (round 8). It did not fully evade, because the classifier weighs 12 features jointly.
- **Novelty of the evasion axis, not the size of the change, predicts difficulty.** Round 15's larger memory reduction than round 8 was caught more confidently, because the classifier had by then seen more memory-focused examples (§2.2, §2.5).

- **Out-of-distribution inputs produce meaningless confidence scores in either direction.** Four rounds (2, 10, 11, 14) pushed mean power outside the reference band; three were flagged as such at the time, one was not (§4). This is the most transferable methodological lesson of the campaign: check whether an input is still inside the region a model was trained to judge before trusting its confidence score, regardless of whether that confidence reads high or low.

7. Limitations

All 16 rounds use a small synthetic transformer (~85–100M parameters), not a pretrained model. Every round is a single execution; only the false-positive investigation's boundary point (outside the scope of this report — see docs/REDTEAM.md) received a statistical-replication check. Round 14 produced an inconclusive result rather than a clean pass or fail. The 239–351 W reference band derives from one baseline run's five categories, not a swept characterization of the full input space this classifier could plausibly encounter.

References

Rahman, R. & Tajdari, S. (2026). *Detecting Hidden ML Training With Zero-Overhead Telemetry.* TAIGR @ ICML 2026. arXiv:2606.19262.

Seferis, E. & Fist, T. (2026). *Detecting Compute Structuring in AI Governance is Likely Feasible.* AAAI-26. doi:10.1609/aaai.v40i44.41127.

Douillard, A. et al. (2023). *DiLoCo: Distributed Low-Communication Training of Language Models.* arXiv:2311.08105.

Full round-by-round narrative, citations, and honest limitations: `docs/REDTEAM.md`. General-audience writeup of the whole campaign (this report plus the separate false-positive investigation): `docs/WRITEUP.md`.